# Toporia library: original-reference audit

Integration branch | 27 October 2026 | Current total: 114 implementations, comprising 88 selectable plugins, nine physics engines and 17 presets. Earlier update pages retain their historical counts.

## Independent MATLAB checks

Unmodified author-distributed top88 and top3d functions were run in MATLAB R2026a Update 5. Toporia matches all three stopping iteration counts and first/last reported objectives to the originals' four-decimal printed precision. Both top88 final mean densities also match within 1e-10. These are pre-update printed objectives, not fresh final-design solves.

| Original command | Iterations | First C | Last C | Volume |
|---|---|---|---|---|
| top88(24,8,.5,3,1.5,1) | 39 | 975.8323 | 265.3387 | .500 |
| top88(24,8,.5,3,1.5,2) | 56 | 975.8323 | 274.4604 | .500 |
| top3d(12,4,2,.3,3,1.5) | 39 | 18934.5908 | 7549.3805 | .300 |

Top3d uses equal nodal depth forces: the Python reference recipe explicitly uses those shares and total tip force 3. Production trapezoidal line-load quadrature is unchanged. Source hashes, commands, observations, runtime and date are retained in tests/references/matlab.json, separately from unchanged internal golden baselines.

## Reproducibility and a qualified RBF reference

Four external-fixture tests pass. The native rerun driver reproduces all three cases. It verifies supplied source hashes, isolates temporary copies and hides figures. Use python -m experiments.original_references; add --matlab --source-dir PATH for original runs. It installs nothing, saves no MATLAB path and never regenerates fixtures automatically.

The publisher's original TOPRBF(40,20,0.5) also completes in 88 iterations, with C=594277.942780 and volume=0.499728 under its -100 tip force. This is a standalone reference. Toporia uses mean rather than median energy normalisation, guarded rescaling and field clipping; sampling already gives a small initial discrepancy. Its earlier faithful-port/authorship wording is corrected without changing the existing algorithm. A 24x8 original attempt lost finite field values and became empty; it is rejected as a valid reference.

## Remaining source and compatibility gaps

Top99neo/top3D125 require absent fsparse and no MATLAB C compiler is selected. MMC188 requires absent mmasub/subsolv. Original TOBS101 stops with the R2026a error: Option CutGeneration for intlinprog has been removed. Challis' original source download returned a challenge page; the reaction-diffusion author's source host was unavailable. Existing local original/helper files can unblock further work. No missing runtime, compiler or toolbox is installed or reconfigured without approval.

Full scope and recipes: experiments/README.md and docs/INTEGRATION-progress.md. Original-code equality applies only to the three strict cases above. Sources, traces, builders and generated runs are removed; retained numbers are test fixtures.

Primary source pages: DTU top88; Liu and Tovar top3d. Original source URLs and SHA-256 hashes are recorded in the reference fixture.

# Toporia library: Hashin-Shtrikman update

Integration branch | 27 October 2026 | 114 implementations: 88 selectable plugins, nine physics engines and 17 presets. Twelve models x 23 updaters plus three whole methods give 279 method names; compatibility still applies.

## Elastic upper envelope at the scenario Poisson ratio

The new hashin_shtrikman law binds the scenario's actual Poisson ratio and dimension. It takes the stricter of the elastic bulk and shear HS upper limits along the solver's fixed-nu ray. For either channel M, q_M=(M0-Mmin)/(Mmin+a_M); q=max(q_bulk,q_shear); E(rho)=Emin+(E0-Emin)*rho/(1+q*(1-rho)). The finite soft phase participates in the bound, rather than being added on top of the zero-stiffness-void formula.

Plane stress uses k=E/[2(1-nu)], g=E/[2(1+nu)], a_k=g and a_g=k*g/(k+2*g). Solid 3-D elasticity uses k=E/[3(1-2*nu)], a_k=4*g/3 and a_g=g*(9*k+8*g)/(6*(k+2*g)). Shifts use the solid phase. Stiffness derivatives are analytic, and the volume budget stays geometric. Original Q4/H8 numerical sources are unchanged.

## Scope and integration

This is a fixed-nu upper envelope, not the density-varying Poisson ratio in Bendsoe and Sigmund equation 17. The plane-stress nu=1/3, Emin=0 case reduces to E0*rho/(3-2*rho). Both upper-channel limits are enforced. Lower bounds, joint attainability, arbitrary phase Poisson ratios and microstructure realisability are not certified. A finite soft phase and extreme nu can put a fixed-nu tensor outside a lower-channel bound.

Optional setup and compatibility hooks leave ordinary laws unchanged. Thermal pairings are refused in preflight and at direct model initialisation: elastic bulk/shear limits are not conductivity limits. Direct Python users must call law.setup(problem). The accepted range is -1 < nu < 0.5, 0 <= Emin <= E0, with E0 positive. There is no separate penalty knob.

## Verification and bounded benchmarks

Seventy-seven focused cases cover independent reciprocal-form HS bounds across six nu values, five stiffness contrasts and both dimensions, plus physical/filter adjoints, context binding, the special case and thermal preflight. The focused/interpolation/unchanged-golden group passes 106 tests. Wider schema/study and GUI checks also exercise the new law.

At nu=0.3, Emin=1e-9, density radius 1.5 and 300 MMA steps (move=0.05), MBB compliance improves from 513.19396 to 199.62258, with volume 0.499988 against 0.5. Cantilever improves from 222.04768 to 83.15603, with volume 0.399769 against 0.4. Gray fractions remain 34.83% / 26.72%; final changes are 0.002999 / 0.050000.

Both runs reach the iteration cap. These are improvement and budget checks, not convergence, binary-design or fabrication claims. The comparison study's common SIMP p=3 referee uses a different material interpolation. Full measurements and limits are recorded in experiments/README.md.

Choose Hashin-Shtrikman upper envelope (fixed nu) in Material law, or interpolation.type=hashin_shtrikman. Reusable no-output recipe: python -m experiments.material_laws --law hashin_shtrikman.

Primary sources: Bendsoe and Sigmund (1999), sections 3.2 and 3.4; Berryman (2005), 3-D HS formulas. Earlier integration updates and linked catalogue follow.

# Toporia library: SINH update

Integration branch | 27 October 2026 | 113 implementations: 87 selectable plugins, nine physics engines and 17 presets.

## Linear stiffness, penalised effective volume

Basic SINH now supplies two material measures: eta1=rho for stiffness and eta2=1-sinh(p*(1-rho))/sinh(p) for effective volume. The p=0 limit is linear in both. At intermediate density eta2 exceeds rho, taxing gray material through the budget rather than penalising its stiffness. The default p=3 can be continued through interpolation.penal.

The implemented variant uses the same density field in both measures: Bruns Table 2 case 5 with shared density filtering, or case 4 without filters. The resource-only filtered case 6 and hybrid stiffness penalty are not implemented. Shared filtering can retain diffuse gray boundaries; this does not reproduce the paper case-6 results.

## Budget and adjoint integration

Identity volume hooks preserve the other material laws. Assembled models apply the selected effective cost in evaluations, cheap updater bisections and volume objectives. Its exact derivative is carried through filters and design representations, in 2-D/3-D. Element-density starts are calibrated to the effective budget; other representations rely on the updater. Passive solids cost one and voids cost zero.

Geometric density remains the displayed field and usual volume history/study column. effective_volume_fraction records the current design's budget cost separately. Limit checks use it and label the effective-volume budget explicitly. Compare Methods adds effective_volume while keeping its common SIMP p=3 referee; the referee and SINH linear-material objective measure different material models. Effective cost is not geometric mass or a homogenisation certificate.

## Bounded validation and measured limits

Twenty-four focused cases cover the material equations, 2-D/3-D physical/filter/network adjoints, five optimiser paths, continuation, variants, pinned material, volume objectives and limit reporting. The focused/interpolation/unchanged-golden group passes 53 tests. Comparison-study and GUI reachability checks also exercise the new law.

With 300 MMA steps, p=3, move=0.05, density radius 1.5 and 60-cell-wide grids, MBB compliance improves from 550.63805 to 202.79554. Its geometric fraction is 0.414981 and effective cost 0.500000 against limit 0.5. Cantilever improves from 234.23086 to 84.58895, with geometric fraction 0.337000 and effective cost 0.398823 against 0.4.

Both runs stop at the cap. Gray fractions are 33.75% / 27.22%, with final changes 0.007759 / 0.038767. These are bounded improvement and effective-budget checks, not convergence, binary designs or direct performance comparisons to penalised SIMP. Fresh final responses and full scope are recorded in experiments/README.md.

Use Material law: SINH (linear stiffness, penalised volume), or interpolation.type=sinh. Reusable no-output recipe: python -m experiments.material_laws --iterations 300 --mesh 1.

Primary source: Bruns (2005), equations 22-23 and Table 2; readable paper. Earlier integration updates and linked catalogue follow.

# Toporia library: neural field update

Integration branch | 26 October 2026 | 112 implementations: 86 selectable plugins, nine physics engines and 17 presets.

## A coordinate network as the design representation

The new neural representation maps normalised physical coordinates to density through dense tanh hidden layers and a sigmoid output. Every weight and bias is an optimisation variable. Explicit NumPy reverse propagation carries the exact physical/filter sensitivities back to these variables. It supports 2-D/3-D and needs no neural runtime, dataset or pretrained weights.

The default network has two hidden layers of width 20: 501 parameters in 2-D, independent of mesh size. Local seeded initialisation preserves reproducibility without altering the global random generator. Output weights start at zero; a logit bias gives the requested uniform starting volume. Finite symmetric weight/bias bounds and initial-volume checks are explicit.

## Independent TOuNN-style variant

This is an independent smooth variant, not the original PyTorch implementation. Batch-normalised LeakyReLU, the density offset, AMSGrad, Fourier features and built-in symmetry/manufacturing rules are absent. No GPL source code was copied. Existing Adam uses an inequality augmented Lagrangian; general-vector MMA also works. Element-specific OC/BESO/SiMPL updaters are refused before running.

## Measured behaviour and limits

On 600-step, seed-0 MBB / cantilever runs, Adam improves compliance from 984.55480 / 609.61732 to 389.76771 / 192.14204, at volumes 0.499926 / 0.399894 (limits 0.5 / 0.4). MMA reaches 346.15640 / 144.21322 at volumes 0.499657 / 0.399825. Both use the same SIMP p=3 and density radius 1.5. Adam lr is 0.01 with decay 0.995; MMA move is 0.005.

These runs stop at the iteration cap. Gray fractions remain 83.33% / 64.89% for Adam and 82.33% / 38.00% for MMA. This is functional integration and bounded improvement, not convergence, crispness, certified length scale or reproduction of reference TOuNN performance. Small weight movement alone cannot establish an adequate design.

Sixteen focused tests cover all weight/bias derivatives in 2-D/3-D, the filtered FE chain, deterministic starts, bounds, mesh-independent parameter count and updater preflight. Forty combined representation tests and 53 selected GUI form/reachability cases pass. Existing schema/composition/golden regressions are also checked.

Use the GUI Design representation panel, or representation.type=neural with q4+adam (lr=0.01) or q4+mma (move=0.005). Reusable recipe: experiments/neural.py; full results and scope are in the representation and experiments READMEs.

Source: Chandrasekhar and Suresh (2021), TOuNN. Reference repository. Earlier integration updates and the linked catalogue follow.

# Toporia library: multiresolution update

Integration branch | 26 October 2026

111 implementations: 85 selectable plugins, nine physics engines and 17 presets. Twelve models and 23 updaters form 276 compositions; three whole methods bring the method-name total to 279.

## Nested Q4 analysis with fine density

The new q4_multiresolution model keeps the design grid fine and solves a nested coarse bilinear displacement space. Stiffness uses exact 2 x 2 Gauss integration in each constant-density fine cell. Fine loads transfer by virtual work; fine supports become interpolation constraints. Complete compliance gradients include all fine-cell contributions. Factor 1 retains the unchanged Q4 reference path.

Scope: aligned, divisible 2-D Cartesian meshes and elastic compliance. This is an independent MTOP variant, not the original centroid-rule implementation, an adaptive/high-order method or an independent three-mesh system. Stress, springs and 3-D are outside this model.

## Bounded benchmark: 100 OC steps

| Preset / factor | System DOFs | Coarse C | Fresh fine C | Volume / limit |
|---|---|---|---|---|
| MBB / 1 | 2540 | 249.35203 | 249.35203 | 0.499996 / 0.5 |
| MBB / 2 | 670 | 244.09112 | 249.59705 | 0.500019 / 0.5 |
| Cantilever / 1 | 3720 | 110.01908 | 110.01908 | 0.399984 / 0.4 |
| Cantilever / 2 | 960 | 108.08059 | 110.14153 | 0.399985 / 0.4 |

Factor 2 underestimates fine compliance by 2.21% / 1.87%. Single-run optimisation timings were 2.125 / 1.099 s (MBB factors 1 / 2) and 3.884 / 1.675 s (cantilever); these illustrate one machine/run, not general speed guarantees. Runs are unconverged. MBB factor 2 has a small OC volume overshoot.

## Verification and known limits

The broad model/composition/schema/golden group passed 987 tests and selected GUI checks passed 102. Eighteen focused cases include independent subcell quadrature, rectangular cells, multiple loads, off-coarse supports and finite differences. A disconnected fine-grid cut deliberately exposes artificially stiff coarse designs. Filtering does not certify removal of QR-patterns. Compare Methods keeps its independent fine-Q4 referee.

Reusable recipe: experiments/multiresolution.py. Implementation and limits: toporia/plugins/models/README.md and physics/README.md.

Sources: Nguyen et al. (2010), MTOP; Gupta et al. (2018), QR-pattern artefacts. Earlier update pages and the linked catalogue follow.

# Projection-based self-support

Gaynor–Guest / independent 2-D cell discretisation

**Current catalogue: 84 selectable plugins + 8 physics engines + 17 presets = 109 implementations.** The method-name total remains 256. This update supersedes earlier totals and remaining-work lists in the following snapshots.

## Method and use

The **self_support** filter multiplies the design field by a projected average of the supported field inside a lower wedge, then applies a local cone average and exponential projection. An exact reverse sweep carries sensitivities through every lower-layer dependency. Both projection sharpnesses support continuation.

Select **Gaynor–Guest self-support projection (2-D)** as the last filter. It includes local regularisation. Use element densities, a rectangular 2-D box and no forced solid/void regions. The first cell layer touches the build plate; +y builds upwards. The angle is measured from the plate, so 90 degrees permits vertical supports only.

**Scope:** cell-centred points replace the paper's nodal examples. The support threshold depends on the discrete grid and angle. Finite smoothing and boundary truncation do not certify an exact angle, minimum feature size or printing process.

## Validation

434 manufacturing/schema/variant checks, 53 GUI checks and 16 unchanged golden cases pass. The preceding full suite after machining passed 2363 tests with one optional multigrid skip. Small move limits down to 0.00001 now survive GUI/schema round trips; the existing default stays 0.2.

The following 500-iteration GCMMA runs use RAMP q 10 to 30 and local sharpness 10 to 25, on better-resolved meshes. Initial and final compliance use the same final material law. Neither run is claimed to have converged.

| Benchmark | Initial to final compliance | Physical volume / limit |
|---|---|---|
| MBB / 60 x 20 | 12522.25648 to 454.66344 | 0.499995 / 0.5 |
| Cantilever / 60 x 30 | 2901.78844 to 296.54728 | 0.399999 / 0.4 |

Final changes remain 0.02508 / 0.01052 and gray fractions 20.50% / 15.83%. Thresholding at 0.5 raises volumes to 0.519167 / 0.411111, above budget. An independent one-layer 45-degree check finds 0 / 1 unsupported cells. Binary designs are not labelled feasible or printable; the report also records their fresh compliance.

## Reproduce and inspect

Run **python -m experiments.self_support --iterations 500 --mesh 1 --updater pymoto_gcmma** (optional pyMOTO). The default script also reproduces the own-MMA small-move trial. Progress and measurements go to the console; no saved run outputs are generated. The README records oscillation, grayness and thresholding limits.

Gaynor & Guest, SMO 54 (2016) 1157–1172 / implementation. New source files need the local integration checkout until pushed. No published code is copied.

Next: multiresolution meshes, TOuNN, SINH / Hashin–Shtrikman laws, and original-code references where a runtime is available. Machining is committed as 4decae8. Each addition remains separate; GUI and launcher source checkouts are unchanged.

# Machining accessibility

26 October 2026 / integration/physics

**Current catalogue: 83 selectable plugins + 8 physics engines + 17 presets = 108 implementations.** Eleven models and 23 updaters form 253 compositions; with three whole methods, there are 256 method names.

## New method: PDE machining

The **machining** filter is an independent Cartesian variant of the Høghøj–Träff advection–diffusion formulation. It intersects material shadows for selected tool approach sides in 2-D or 3-D and provides the complete gradient through the PDE, intersection and projection. Passive solids cast shadows.

Use **Element densities**, a density/PDE regulariser, then **PDE machining (axis directions)** as the last filter. A tool from +y approaches from above. Forced-void regions are refused. Initial source densities meet the physical volume budget; convergence also monitors physical-density changes.

**Recipe:** MMA; move 0.01; initial asymptotes 0.03; growth 1.05; shrink 0.65; tolerance 0.00001. Defaults: Peclet 10000, projection beta 8, threshold 0.4 and shadow power -4. Large optimiser moves can be unstable.

## Validation and scope

Twenty-five focused numerical tests pass. The 80-iteration coarse-grid runs below reevaluate the final design; neither is claimed to have converged. Independent 0.5-threshold ray checks found zero inaccessible void cells for the selected tool side.

| Benchmark | Compliance: initial to final | Physical volume |
|---|---|---|
| MBB / 24 x 8 | 217476025.72 to 686.42380 | 0.499864 / 0.5 |
| 3-D cantilever / 24 x 8 x 4 | 648.33247 to 76.06831 | 0.299961 / 0.3 |

**Limits:** Cartesian directions and ideal zero-width tools. No finite cutter, holder collision or minimum-size guarantee. Diffusion, projection and gray densities can leave accessibility artefacts. This is not a port of Langelaar's finite-tool method or a machining certificate.

## Other integration fixes

Small mechanism loads retain their precision. Compare Methods selects compatible physics and objectives; all 17 presets have GUI-to-study regressions, plus a 321-combination compatible-updater smoke audit. These short audits check execution, not convergence.

## Source and continuation

Høghøj & Träff, CMAME 391 (2022) 114488 / readable full paper. Local code: machining.py (new files need the local checkout until pushed). Reproduce: python -m experiments.machining --iterations 80. The script prints measurements without saving runs. Full regression: 2363 tests passed; one optional multigrid check skipped.

Next: Gaynor–Guest self-support; multiresolution meshes; TOuNN dependency check and integration; SINH / Hashin–Shtrikman laws; original-code references where a runtime is available. Each addition is completed and committed separately.

This update takes precedence over the earlier 13-page catalogue snapshot that follows (checkpoint 9abc824). Its completed items and historical sources are preserved. GUI and launcher source checkouts remain separate. No code is pushed or merged.

# The library

## What is implemented, what changed, and what remains

A linked catalogue of the current integration checkout, updated with the work completed through minimum-length constraints.

| 82 | 8 | 17 |
|---|---|---|
| selectable plugins | physics engines | benchmark presets |

107 implementations in total. The 11 models and 23 updaters form 253 named compositions, plus three whole methods: 256 method names.

**26 October 2026**
Branch `integration/physics` / implementation checkpoint `9abc824`

Completed work / Current catalogue / Remaining work

Work is isolated in the integration checkout. The GUI and launcher source checkouts are unchanged; the owner reviews and merges later.

Paper and repository links are clickable. Source-code links target integration/physics; newly committed files may require the local checkout until the branch is pushed. This document does not publish or merge code.

# Completed integration work

The small and medium lists from the original PDF define this round. Existing physics was stabilised and checked before new methods were added. Each row below has its own commit; restricted variants are identified explicitly.

| Addition | Implemented scope / result | Commit |
|---|---|---|
| Existing physics stabilised | Repaired compatible compositions, scaling, constraint guards and numerical issues. Existing thermal, dynamic and nonlinear physics remain in the catalogue. | 2dc42a1 |
| Q4 linearised buckling | Recovered and validated the eigenproblem, complete adjoint and column benchmark; sparse solution and deterministic mode handling. | b9f7a7f |
| Anisotropic Helmholtz filter | Conservative 2-D/3-D finite-volume filter with directional radii, preserved constants/volume and an exact adjoint. | ce9f4e2 |
| Moving morphable voids | Restricted 2-D superelliptical voids through the MMC kernel. No 3-D B-spline surfaces or FE pruning. | 6a10f2b |
| Floating projection | Updater with mass-preserving threshold, fixed-region/bound handling and internal beta continuation. The original backlog's 'without beta continuation' description is corrected. | 00ffa9b |
| Annealing evaluation | Own bounded standard-SA experiment; not an adaptive topASA port. It missed the unchanged 10%-of-OC gate on three seeds at 200 and 2000 solves and is not registered. | 8116acf |

These are independently implemented paper variants unless the catalogue explicitly identifies a port or an external-package wrapper. No original code was copied where reuse permission was unclear.

# Completed work / continued

| Addition | Implemented scope / result | Commit |
|---|---|---|
| Compliant mechanisms | Physical input/output ground springs, signed output-displacement adjoint, a dedicated Q4 model and Force Inverter preset; pyMOTO cross-check. | 4496e90 |
| Geometry projection | Independent 2-D/3-D fixed-radius capsule bars with endpoint and activation variables, bounded union and complete derivatives; no GPTO code copied. | 55c019e |
| Reaction-diffusion level set | Sparse cell-centred 2-D compliance method using the paper's elastic topological-sensitivity tensor, an implicit PDE step and updated-volume control. | c76509f |
| Hamilton-Jacobi level set | Godunov evolution with circular-hole plane-stress sensitivity, hole nucleation, reinitialisation and volume feedback. Not a standalone Amstutz geodesic solver. | a27b13b |
| Phase-field energy | 2-D/3-D regularised-compliance objective with a physical diffuse-interface energy and exact derivatives. Uses existing optimisers; no dedicated Allen-Cahn/Cahn-Hilliard solver. | 71abd18 |
| Geometric minimum length | Solid and void constraints through a separate cone filter, complete adjoint, conservative nodal KS and fixed 1-D tolerance calibration. Not an exact geometric size certificate. | 9abc824 |

## Validation checkpoint

The latest full native run took 28 min 13 sec: **2303 passed, one optional multigrid skip and one outdated capability assertion failed**. Only that expected capability tuple was corrected; all six tests in its file passed on rerun. All **2304 current tests have passing coverage** across those runs. The original full-run log is retained, rather than described as a clean monolithic run.

The new methods have gradient, physical/geometric fixture and composed-run checks. GUI reachability, original golden checks, Ruff and whitespace checks pass. Original Q4/H8 sources and golden baselines are unchanged. MATLAB/Octave reference runs remain optional and have not been claimed as completed.

**Minimum-length limitation:** uniform gray starts stalled or retained violations in tested strict two-phase runs. A resolved feasible-seed cantilever with linear ersatz improved compliance from 115.115877 to 111.907874 at volume 0.5; fresh solid/void residuals were -2.22e-7 / -3.23e-8. This is a modest local validation, not evidence of reliable topology creation from a uniform start.

Detailed integration record / Reproducible validation recipes

# The current catalogue

The following tables are generated from `toporia/plugins/LIBRARY.md`. They include the new entries and the already implemented physics. Source and paper references retain their clickable links. Optional packages remain separate dependencies.

**Port**: translated from named code. **Paper**: independently written from a paper. **Wraps**: calls an optional external package. **Own**: Toporia's own implementation or standard practice.

### Design representations: what the design variables are (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Element densities | `element_density` | element_density.py | Own | Bendsøe 1989; Sigmund's 99-line code · papers 1 2 |
| Moving morphable components (constant or variable thickness) | `mmc` | mmc.py | Paper | Guo, Zhang & Zhong 2014; variable thickness as in MMC188 (Zhang et al., *SMO* 2016) · papers 1 2 |
| Moving morphable voids (restricted 2D superellipses) | `mmv` | mmv.py | Paper | Zhang et al., *CMAME* 2017; own geometric dual of MMC, without B-spline surfaces or FE pruning · paper |
| Geometry projection (2D/3D capsule bars) | `geometry_projection` | geometry_projection.py | Paper | Own fixed-radius, bounded-union variant with component activation; no GPTO code copied · papers 1 2 |
| B-spline field | `bspline` | bspline.py | Paper | Qian, *CMAME* 2013 · paper |

## Filters: design to physical density, and fabrication rules (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Density filter | `density` | density.py | Port | top88 (Andreassen et al. 2011); Bruns & Tortorelli 2001; Bourdin 2001 · papers 1 2 3 |
| PDE (Helmholtz) filter | `pde` | pde.py | Port | the PDE variant of top88 (top82.m); Lazarov & Sigmund 2011 · papers 1 2 |
| Anisotropic PDE (Helmholtz) filter | `pde_anisotropic` | pde_anisotropic.py | Paper | Directional extension of Lazarov & Sigmund 2011, using conservative cell-centred finite volumes · paper |
| Sensitivity filter | `sensitivity` | sensitivity.py | Port | Sigmund 1997, as in the 99-line code and top88 · papers 1 2 3 |
| Heaviside projection (fixed or volume-preserving threshold) | `heaviside` | heaviside.py | Paper | Guest et al. 2004; Wang, Lazarov & Sigmund 2011; volume-preserving as in top99neo (Ferrari & Sigmund 2020) · papers 1 2 3 |
| Morphology filter (erode / dilate / open / close) | `morphology` | morphology.py | Paper | Sigmund, *SMO* 2007 · paper |
| AM overhang filter | `am` | am.py | Port | Langelaar's AMfilter.m (*SMO* 2017) · papers 1 2 |
| Routing radius filter | `routing` | routing.py | Own | inspired by Sigmund 2007 · paper |
| Symmetry filter | `symmetry` | symmetry.py | Own | standard practice · paper |
| Extrusion (constant cross-section) | `extrusion` | extrusion.py | Paper | Ishii & Aomura 2004 · paper |
| Pattern repetition | `pattern` | pattern.py | Paper | Bendsøe & Sigmund 2003; Almeida, Paulino & Silva 2009 · papers 1 2 |
| Casting (no undercuts) | `casting` | casting.py | Paper | Gersborg & Andreasen, *SMO* 2011 · paper |

## Material laws: density to stiffness (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| SIMP (power law) | `simp` | simp.py | Paper | Bendsøe 1989 · paper |
| RAMP (rational) | `ramp` | ramp.py | Paper | Stolpe & Svanberg 2001 · paper |
| Ordered SIMP (multi-material) | `ordered_simp` | ordered_simp.py | Paper | Zuo & Saitou, *SMO* 2017 · paper |

## Physics models: what computes displacements and gradients (folder, elements in physics/)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| 2-D Q4 plane stress (Toporia) | `q4` | q4.py, q4_plane_stress.py | Port | the physics half of top88 · paper |
| 2-D Q4 compliant mechanism (Toporia) | `q4_mechanism` | q4_mechanism.py, q4_mechanism.py | Paper | Own port-spring Q4 engine; optional independent pyMOTO assembly/backpropagation check · paper |
| 2-D Q4 with buckling (Toporia) | `q4_buckling` | q4_buckling.py, q4_buckling.py | Paper | Ferrari, Sigmund & Guest 2021; Toporia's p-norm aggregate and column geometry · paper |
| 2-D Q4, large deformation (Toporia) | `q4_nonlinear` | q4_nonlinear.py, q4_nonlinear.py | Paper | Buhl, Pedersen & Sigmund, *SMO* 2000; energy interpolation after Wang, Lazarov, Sigmund & Jensen, *CMAME* 2014; open reference: Han et al. 2021 · papers 1 2 3 |
| 3-D H8 bricks (Toporia) | `h8` | h8.py, h8_solid.py | Port | top3d (Liu & Tovar, *SMO* 2014) · paper |
| Linear elasticity, pyMOTO network | `pymoto_elastic` | pymoto_elastic.py | Wraps | pyMOTO (MIT), `pip install pymoto` |
| 2-D Q4 heat conduction (Toporia) | `q4_heat` | q4_heat.py, q4_heat.py | Paper | Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al., *SMO* 2006 · paper |
| 3-D H8 heat conduction (Toporia) | `h8_heat` | h8_heat.py, h8_heat.py | Paper | the 3-D brick of the same heat problem · paper |
| Heat conduction, pyMOTO network | `pymoto_heat` | pymoto_heat.py | Wraps | pyMOTO's `AssemblePoisson` (MIT), `pip install pymoto`; the cross-check of `q4_heat` |
| 2-D Q4 with vibration (Toporia) | `q4_dynamic` | q4_dynamic.py, q4_dynamic.py | Paper | Pedersen, *SMO* 2000; Du & Olhoff, *SMO* 2007 · papers 1 2 |
| Vibration, pyMOTO network | `pymoto_dynamic` | pymoto_dynamic.py | Wraps | pyMOTO `AssembleMass` and `EigenSolve`, `pip install pymoto` |

## Objectives and constraints (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Compliance | `compliance` | compliance.py | Own | the classic objective (Bendsøe & Sigmund 2003) · paper |
| Phase-field regularised compliance | `phase_field` | phase_field.py | Paper | Own 2-D/3-D fixed-load diffuse-interface energy, with existing gradient optimisers; no separate PDE solver · papers 1 2 |
| Output displacement (maximise) | `output_displacement` | output_displacement.py | Paper | Signed point output, one shared spring-augmented adjoint · paper |
| Material volume | `volume` | volume.py | Own | as in Le et al. 2010 and pyMOTO's stress example · paper |
| Geometric minimum feature length | `minimum_length` | minimum_length.py | Paper | Own 2-D/3-D physical-field variant, separate cone filter, full indicator adjoint, conservative nodal KS and fixed tolerance · papers 1 2 |
| Local von Mises stress (augmented Lagrangian) | `local_stress` | local_stress.py | Paper | Giraldo-Londoño & Paulino, *SMO* 2021 (PolyStress); Senhora et al., *SMO* 2020 · papers 1 2 |
| End compliance (non-linear) | `end_compliance` | end_compliance.py | Paper | Buhl, Pedersen & Sigmund, *SMO* 2000 · paper |
| Peak von Mises stress | `stress` | stress.py | Paper | Le, Norato, Bruns, Ha & Tortorelli, *SMO* 2010 · paper |
| Buckling load factor | `buckling_load` | buckling.py | Paper | Reciprocal-factor aggregate; objective or constraint on `q4_buckling` · paper |
| Thermal compliance | `thermal_compliance` | thermal_compliance.py | Paper | Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al., *SMO* 2006 · paper |
| Fundamental frequency (maximise) | `eigenfrequency` | eigenfrequency.py | Paper | Díaz & Kikuchi, *IJNME* 1992; Du & Olhoff, *SMO* 2007 · papers 1 2 |
| Fundamental frequency (at least) | `eigenfrequency_limit` | eigenfrequency.py | Paper | the same, as a constraint |
| Dynamic compliance (harmonic load) | `dynamic_compliance` | dynamic_compliance.py | Paper | Ma, Kikuchi & Hagiwara, *Comput. Mech.* 1993 (no link); PolyDyna · paper |

## Optimisers: how the design moves (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Optimality criteria | `oc` | oc.py | Port | the 99-line code and top88 · papers 1 2 |
| Floating projection (FPTO) | `floating_projection` | floating_projection.py | Paper | Huang, *Engineering Structures* 2020 and *Adv. Eng. Softw.* 2021; modular updater with internal β continuation · papers 1 2 |
| Optimality criteria, several constraints | `oc_multi` | oc_multi.py | Own | after Zhou & Rozvany 1991 · paper |
| Method of moving asymptotes | `mma` | mma.py | Port | Svanberg 1987, one-constraint form of top88's MMA variant · papers 1 2 |
| CONLIN (convex linearisation) | `conlin` | conlin.py | Paper | Fleury & Braibant 1986; Fleury 1989 · papers 1 2 |
| Mirror descent (SiMPL) | `simpl` | simpl.py | Paper | Keith, Kim, Lazarov & Surowiec, *SMO* 2025 · paper |
| BESO (soft kill) | `beso` | beso.py | Paper | Huang & Xie 2007, 2010 · papers 1 2 |
| BESO (hard kill) | `beso_hard` | beso_hard.py | Paper | Huang & Xie 2010 · papers 1 2 |
| TOBS (integer LP) | `tobs` | tobs.py | Paper | Sivapuram & Picelli 2018; the 101-line code of Picelli et al. 2021; solved with HiGHS (SciPy) · papers 1 2 |
| Augmented Lagrangian (local constraints) | `augmented_lagrangian` | augmented_lagrangian.py | Paper | PolyStress outer loop (Giraldo-Londoño & Paulino 2021) with MMA inner steps · papers 1 2 |
| Adam (augmented Lagrangian) | `adam` | adam.py | Paper | Kingma & Ba 2015; the volume penalty of TOuNN (Chandrasekhar & Suresh 2021) · papers 1 2 |
| Proportional topology optimisation | `pto` | pto.py | Paper | Biyikli & To, *PLoS ONE* 2015 · paper |
| MMA (pyMOTO), GCMMA (pyMOTO) | `pymoto_mma`, `pymoto_gcmma` | pymoto_optimizers.py | Wraps | pyMOTO (MIT) · papers 1 2 |
| MMA (mmapy), GCMMA (mmapy) | `mmapy_mma`, `mmapy_gcmma` | mmapy_optimizers.py | Wraps | mmapy, Svanberg's reference code in Python (**GPL-3**, optional), `pip install mmapy` · papers 1 2 |
| SLSQP (SciPy) | `scipy_slsqp` | scipy_slsqp.py | Wraps | SciPy (BSD), Kraft 1988 |
| trust-constr (SciPy) | `scipy_trust_constr` | scipy_minimize.py | Wraps | SciPy (BSD) · paper |
| MMA, CCSAQ, SLSQP, augmented Lagrangian + L-BFGS (NLopt) | `nlopt_mma`, `nlopt_ccsaq`, `nlopt_slsqp`, `nlopt_auglag` | nlopt_optimizers.py | Wraps | NLopt (LGPL/MIT), `pip install nlopt` · papers 1 2 3 4 |
| Null-space optimiser | `nullspace` | nullspace.py | Wraps | null-space optimizer of Feppon, Allaire & Dapogny 2020 (**GPL-3**, optional), `pip install nullspace_optimizer` · paper |

## Whole methods: their own loop (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| RBF level set | `levelset` | levelset_rbf.py | Port | TOPRBF.m (Luo et al.); Wang & Wang, *CMAME* 2006 · papers 1 2 |
| Reaction-diffusion level set | `levelset_reaction_diffusion` | levelset_reaction_diffusion.py | Paper | Otomori et al. 2015; own cell-centred variant, implicit diffusion and bisected volume multiplier · paper |
| Hamilton-Jacobi level set with circular-hole derivative | `levelset_hamilton_jacobi` | levelset_hamilton_jacobi.py | Paper | Challis 2010; independent upwind HJ/TD, reinitialisation and normalised augmented-Lagrangian volume feedback · paper |

Every model can also be paired with every optimiser as a method called `<model>+<optimiser>`, for example `q4+nlopt_mma`. Those pairings are built on demand, so they are not listed here.

## Continuation: parameters that change during a run (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Equal steps | `steps` | steps.py | Own | the classic SIMP continuation (Sigmund & Petersson 1998) · paper |
| Multiply by a factor | `geometric` | geometric.py | Own | the usual β continuation (Guest et al. 2004; Wang et al. 2011) · papers 1 2 |
| Linear ramp | `linear` | linear.py | Own | - |
| When the design settles | `settled` | settled.py | Own | common practice in the projection literature (Wang et al. 2011) · paper |

The robust formulation (one design evaluated in several versions) is a solver setting rather than a schedule. See the schedules README and the preset *MBB Beam (Robust)*.

## Post-processing: what is made of the finished design (folder)

| In the app | Name | File | Came in | From |
|---|---|---|---|---|
| Threshold to black and white | `threshold` | threshold.py | Own | as method comparisons do (Rojas-Labanda & Stolpe 2015) · paper |
| Connectivity check | `connectivity` | connectivity.py | Own | - |
| Feature size | `feature_size` | feature_size.py | Own | measures what Sigmund 2007 and Wang et al. 2011 promise · papers 1 2 |
| Grey level (M_nd) over the run | `grey_level` | grey_level.py | Paper | Sigmund 2007 · paper |
| Export outline (SVG), (DXF) | `export_svg`, `export_dxf` | export_svg.py, export_dxf.py | Own | - |
| Smooth outline (SVG + DXF) | `smooth_outline` | smooth_outline.py | Paper | Subedi et al. 2020 (review); smoothing splines or Chaikin 1974 · papers 1 2 |
| Export solid (STL) | `export_stl` | export_stl.py | Own | - |

## Benchmark problems (folder)

| In the app | File | From |
|---|---|---|
| MBB Beam | mbb_beam.py | the 99-line code and top88 · papers 1 2 |
| MBB Beam (Robust) | mbb_beam_robust.py | Wang, Lazarov & Sigmund 2011 · paper |
| Vibrating Beam | vibrating_beam.py | after Du & Olhoff 2007 · paper |
| Column Buckling | column_buckling.py | Toporia column geometry, Ferrari, Sigmund & Guest's linearised formulation · paper |
| Cantilever | cantilever.py | the 99-line code · paper |
| Nonlinear Cantilever | nonlinear_cantilever.py | Buhl, Pedersen & Sigmund 2000; Wang et al. 2014 · papers 1 2 |
| Cantilever 3D | cantilever_3d.py | top3d (Liu & Tovar 2014) · paper |
| Michell Cantilever | michell_cantilever.py | Michell 1904 · paper |
| L-bracket (local stress) | l_bracket_stress.py | Giraldo-Londoño & Paulino 2021 (PolyStress); geometry of the L-bracket, limit chosen here · paper |
| L-bracket | l_bracket.py | Duysinx & Bendsøe 1998; Le et al. 2010 (dimensions still to be checked against the papers) · papers 1 2 |
| Three-Point Bending | three_point_bending.py | the materials test |
| Heat Sink | heat_sink.py | Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al. 2006 · paper |
| Standard Bar | standard_bar.py | Toporia's own |
| Drone Arm, Drone Arm (Camera View), Drone Arm (Point Loads) | drone_arm.py, drone_arm_camera_view.py, drone_arm_point_loads.py | Toporia's own application case |
| Force Inverter | force_inverter.py | Own 40 × 40 geometry and port values following Sigmund 1997; signed output and input/output ground springs · paper |

## Evaluated but not registered

The bounded simulated-annealing experiment misses the unchanged 10%-of-OC performance gate on all three seeds at 200 and 2000 solves. It remains reproducible outside plugin discovery; no failing part was registered and no threshold was relaxed. This is an own standard-SA baseline, not a port or evaluation of adaptive topASA. Catalogue implementation counts exclude unregistered experiments.

# Remaining work

**Current linear sequence:** machining accessibility, projection-based self-support, multiresolution meshes, then neural reparameterisation (subject to dependency support). Machining has no code changes yet: a readable full-text copy of Langelaar (2019) is needed after institutional-download failures.

The original scope PDF remains unchanged. MATLAB/Octave oracles are optional; SINH/Hashin-Shtrikman and large framework work remain outside this agreed round.

Listed from least to most effort. "Code" means open code exists (port it, or check against it). "Paper" means it would be written from the paper. Details and status are in TODO.md.

## Small: one file in an existing slot

| What | Kind | Source | Notes |
|---|---|---|---|
| Reference results from the original MATLAB codes (top88, top99neo, top3d, TOBS, MMC, the level-set codes) | tests | Code: run once in GNU Octave or MATLAB · papers 1 2 3 4 5 6 7 | proves the Python versions give the same answers |

## Medium: new physics or a whole method

| What | Source |
|---|---|
| Machining accessibility, projection-based self-support | Paper: Langelaar 2019; Gaynor & Guest 2016 · papers 1 2 |
| Multi-resolution design and analysis meshes | Paper: Nguyen et al. 2010 · paper |
| Neural reparameterisation (TOuNN) | Code: TOuNN (needs PyTorch or JAX, which may not install on Python 3.14) · paper |
| SINH and Hashin-Shtrikman material laws | Paper: Bruns 2005; Bendsøe & Sigmund 1999 (the material law has to see the volume and Poisson's ratio) · papers 1 2 |

# Beyond this round

The original long-term research map is retained below. These larger framework changes and runtime blockers were not implemented in this integration round.

## Large: the framework has to change first

| What | Unlocks |
|---|---|
| Several fields per element | DMO, fibre orientation, true multi-material |
| Unstructured meshes | PolyTop/PolyMesher, VEM, FEniTop, CAD import |
| Beam and truss elements | ground structures (GRAND), Michell layouts |
| Multiscale, IGA, CutFEM | de-homogenisation (deHomTop808), IgaTop, GridapTopOpt |
| Fluids, transient, acoustics, photonics | topflow, TOFLUX, ceviche, Meep |
| Learned methods | TopoDiff, TopologyGAN, dl4to and the SELTO datasets |

## Historical runtime blockers

These runtime statements are retained from the original 24 October survey; package availability was not rechecked for this PDF update.

| What | Why |
|---|---|
| IPOPT (`cyipopt`) | no Windows build for Python 3.14; would need conda or an older Python |
| ParOpt | not on PyPI |
| TOFLUX | the repository states no licence, so it cannot be reused |
| MATLAB bridge (run the originals as whole methods) | the MATLAB Engine for R2026a supports Python ≤ 3.13; GNU Octave is the free alternative |

## References retained from the original scope

Amstutz & Andra (2006): standalone geodesic topological-gradient algorithms remain a candidate; the implemented HJ/TD variant is described in the catalogue. NLopt documentation remains the upstream guide for its wrapped optimisers.